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Original article

Effects of Advancing Wellness and Resiliency in Education Grants on Low-Income Children's Mental Health-care Utilization and Outcomes

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A B S T R A C T

Purpose: Federal Advancing Wellness and Resiliency in Education (AWARE) grants have been awarded to states and local educational agencies (LEAs) to aid them in developing sustainable infrastructure for delivering school-based mental health services, but there is little rigorous evidence on their use and effectiveness. We help to close this gap by examining how the introduction of AWARE grants changes in mental health-care utilization, diagnoses, and disciplinary outcomes among Medicaid-enrolled, low-income school-aged children, and how AWARE grants are being used to increase school capacities for serving children's mental health needs.
Methods: Effects of AWARE grants awarded in Tennessee in 2014 and 2019 were estimated using linked health and education data for all Medicaid-enrolled, school-aged children from 2012–2023 on children's mental health-care utilization, mental health diagnoses, and disciplinary outcomes. Difference-in-differences methods were employed to compare grantee school districts to those without grants and school-based health centers. Qualitative data were collected in interviews with grantees and analyzed to illuminate implementation and mechanisms of grant effects.
Results: AWARE grants increased diagnoses of low-income student mental health conditions and decreased outpatient school-based health claims and exclusionary discipline actions.
Discussion: Qualitative insights showing that AWARE grantees hired in-school mental health staff to identify and meet student mental health needs (regardless of health insurance status or families' ability to pay) are consistent with the quantitatively estimated decline in outpatient school-based claims. Diagnoses of low-income student mental health conditions increased, while rates of exclusionary discipline declined, achieving an important grantee goal.

IMPLICATIONS AND CONTRIBUTIONS

Federal Advancing Wellness and Resiliency in Education (AWARE) grants, awarded to states and localities to aid in developing sustainable infrastructure for delivering school-based mental health services, build capacity to identify and serve students with mental health needs and reduce rates of student exclusionary discipline in Tennessee.

Delivering high-quality education and learning opportunities requires investing in interventions and supports that strengthen children's mental health. A growing research base on mental health services in schools points to a range of potential benefits, including increased access to care, particularly for low-income children [1], earlier identification of mental health conditions

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[2], greater participation in and adherence to treatment [3,4], increased teacher competence and reduced classroom behavioral issues [2], decreased stigma [5], and improvements in children's academic outcomes [6].

Primarily through federal funding, school-based health care has expanded with more than 2,300 school-based health centers (SBHCs) nationwide and steadily added mental health services for children. Estimates indicate that approximately 70% of SBHCs offer mental health services through licensed clinical social workers, psychologists, and other personnel [7,8]. Yet with more than 13,000 U.S. public school districts and 95,000 public schools, this implies that about 83% of school districts lack an SBHC to serve children's health needs.

Since 2014, the Substance Abuse and Mental Health Services Administration (SAMHSA) has awarded Advancing Wellness and Resiliency in Education (AWARE) grants to states and local educational agencies (LEAs) to aid them in developing sustainable infrastructure for delivering school-based mental health services. The specific grant objectives are to increase awareness of youth mental health issues, train school personnel to identify and respond to youth mental health needs, and connect youth with behavioral health issues to appropriate services. The grants have generally been limited to 5-year state awards, ranging from $1,800,000 to $1,950,000 per year each, and one-time LEA awards. Recipients are expected to collaborate with community-based providers to implement mental health-related promotion, awareness, prevention, intervention, and resilience activities for school-aged youth.

As of 2024, 752 AWARE grants (totaling $885 million) have been awarded in 49 of 50 states and some territories to increase school capacities for addressing children's mental health needs, yet there is little rigorous evidence on their use and effectiveness in improving student outcomes.[1] Among states that have commissioned external evaluations of their AWARE grants and made them publicly accessible, most are primarily descriptive.[2] For example, a study of AWARE grants in three Washington state LEAs reported that all students received AWARE-funded services, which contributed to improvements in mental health literacy, reductions in stigma, and more diverse, culturally competent school staff [9]. A California Department of Education AWARE grant evaluation described an increase in mental health professionals and school social workers in three LEAs, the opening of a Family Resource Center, and subsequent reductions in suspensions, substance use, and suicide ideation among students [10]. Another study of AWARE funding in three Michigan counties employed a pre–post, 3-month follow-up design with voluntary participants and reported that Youth Mental Health First Aid (YMHFA) trainings provided through the grant were effective for staff without mental health credentials [11]. Only one study we identified—in a West Virginia county—included a comparison group to assess AWARE grants effects [12]. Comparing students (pre and post) in three AWARE schools to those in seven nonproject schools, the analyses indicated that the AWARE funding contributed to reductions in student absences and disciplinary referrals and improvements in grade point averages for some students.

We help to close this gap in the evidence base by addressing the following research questions in a mixed methods study of AWARE grants awarded in Tennessee: (1) To what extent does the introduction of AWARE grants change mental health-care utilization, diagnoses, and disciplinary outcomes among school-aged children (assessed quantitatively), and (2) how are AWARE grants being used to increase school capacities for serving children's mental health needs (investigated qualitatively)?

Tennessee was one of the 20 states to receive the first 5-year AWARE grants in 2014; one of the five states to receive an AWARE grant in 2019, and a recipient of another AWARE grant in 2021. The Tennessee Department of Education allocated the 2014 AWARE funds to three school districts, 2019 AWARE grants to four school districts, and 2021 AWARE funds to three more school districts. The intent was to alleviate significant gaps in access to mental health services and supports in schools, prioritizing economically distressed and rural areas less likely to have resources to develop these capacities. A project AWARE evaluation described the state and local infrastructure for school-based behavioral health services that the AWARE grants have supported, including school and community partnerships and Multi-Tiered Systems of Support (MTSS)[3] customized to each school district's needs [13]. If AWARE grants increase Tennessee school capacities to diagnose and treat students' mental health needs, we would expect to observe increases in student mental health-care utilization and diagnoses and reductions in disciplinary actions. Exclusionary discipline is a widely observed response to children's externalizing behaviors in schools [14]; therefore, improving student behavior through preventive Tier II and III mental health services in schools has the potential to reduce student disciplinary referrals [15].

Methods

We employed a simultaneous mixed-methods approach to understand how AWARE grants affect school district capacities for providing student mental health supports and student health-care utilization and outcomes. We assembled both quantitative and qualitative data that were integrated at the analysis stage to provide rich descriptive and contextual information for interpreting the findings [16].

Study data and samples

With approval from the Vanderbilt University Institutional Review Board (#182125) and using a probabilistic matching algorithm (including name, social security number, and date of birth), we linked health and education data for all low-income (Medicaid-enrolled) school-aged children in Tennessee from 2012 to 2023, who represent that over two-thirds of students enrolled in a Tennessee public school during this period. Mental health-care utilization and health diagnoses measures were

[1] Author calculations from the SAMHSA grant dashboard, data downloaded September 23, 2024: https://www.samhsa.gov/grants/grants-dashboard.

[2] AWARE grant recipients are required to use a uniform data collection tool to assess the number of individuals who have received training and demonstrated improvement in knowledge, attitudes, and beliefs as a result of training; formal inter/intra-organizational agreements to improve mental health-related practices; policy changes completed as a result of the grant; and individuals screened for mental health, referred to services, and receiving services after referral.

[3] The basic MTSS framework includes strategies to promote the social and personal well-being and development of all students (Tier I), strategies to support students at risk of or with mild behavioral health challenges (Tier II), and strategies to support students who need more individualized and intensive supports (Tier III).

Table 1
School characteristics and outcomes in baseline academic year (2012–13)

School-level characteristics and outcomes (%)	All AWARE grantees		2014 grantees		2019 grantees		Comparison group	
	N	Mean	N	Mean	N	Mean	N	Mean
Economically disadvantaged	86	**0.696**	36	**0.603**	33	**0.740**	588	0.517
Black	86	**0.173**	36	**0.120**	33	**0.223**	588	0.083
Hispanic	86	**0.030**	36	**0.017**	33	0.036	588	0.058
Other race	86	**0.009**	36	0.010	33	**0.009**	588	0.023
Female	86	0.476	36	0.480	33	0.473	588	0.487
First language not English	86	**0.017**	36	**0.010**	33	**0.022**	588	0.049
Special educational needs	86	0.163	36	0.169	33	0.158	588	0.158
Middle school	86	0.321	36	0.335	33	0.307	588	0.315
High school	86	0.194	36	0.185	33	0.212	588	0.200
Mental health condition diagnoses[a]	86	0.156	36	0.171	33	0.152	588	0.168
Outpatient mental health visit	86	0.140	36	0.135	33	0.152	588	0.146
Outpatient school-based claim	86	**0.021**	36	**0.009**	33	0.033	588	0.042
Eval/management of mental health diagnosis	86	0.068	36	0.067	33	0.071	588	0.075
Telehealth - mental health	86	**0.007**	36	0.005	33	**0.011**	588	0.005
Psychotherapy - mental health	86	**0.045**	36	0.050	33	**0.035**	588	0.060
Mental health medication use	86	0.139	36	0.156	33	0.133	588	0.152
Any ED visit	86	0.319	36	0.303	33	**0.369**	588	0.308
Expelled from school	86	0.076	36	0.079	33	0.090	588	0.064
In-school suspension	86	0.007	36	0.003	33	0.010	588	0.006
Out-of-school suspension	86	**0.052**	36	0.042	33	**0.067**	588	0.029

[a] This is composite measure includes diagnoses of anxiety, depression, attention deficit and hyperactivity disorder (ADHD), bipolar disorder, and self-harm or suicidal ideation or suicide. Coefficients in boldface indicate treatment-comparison group differences that are statistically significant at $\alpha < 0.05$.

derived from Medicaid insurance claims (see online Appendix A for details on the construction of these measures). Data from Tennessee Department of Education provided children's demographic characteristics and disciplinary outcomes. These data facilitate analysis of low-income, school-aged children's mental health-care utilization, mental health diagnoses, and disciplinary outcomes before and after AWARE grant distribution for treated and comparison school districts. Research links disciplinary actions to student mental health [14] and long-term negative outcomes (e.g., dropout, criminal involvement, and incarceration) [17].

As AWARE grants were awarded to school districts and the interview data and evaluation reports indicate that AWARE funding has typically been used to increase health staffing and supports in schools, we assessed AWARE grant effects at both district and school levels. AWARE grant districts were defined as always treated after the grant start date, as sustaining the mental health services infrastructure beyond the grant end was an explicit goal of the grant program [13]. In addition, because the state did not allocate AWARE grants to school districts that had substantial, existing health infrastructure and resources, we excluded from the non-grantee comparison group districts with SBHCs before the grant start. In fact, AWARE grantees frequently aspired to develop SBHCs, as did one 2014 grant recipient during the final grant year [18]. Therefore, our primary analytic sample consists of all AWARE-treated districts and others in the state as comparison districts, excluding any that already had established SBHCs.

In the analyses, we estimated combined AWARE grant effects but also distinguished effects for 2014 grantees from those of 2019 grantees, both of which we observed for 4 years after grant receipt, including during the COVID-19 pandemic for the 2019 grantees.[4] Table 1 presents descriptive statistics for treatment and comparison schools in the study sample in the baseline (2012–2013) academic year (before the first AWARE grant award). The first three columns provide school-level descriptive statistics for the 2014 and 2019 AWARE grantees combined and separately, and comparison group statistics are presented in the final column. Rows with lighter shading are sub-measures of the darker shaded measures above them. Results of t-tests comparing the AWARE grantee and comparison group means are indicated in boldface if the differences were statistically significant. Schools in AWARE grantee districts had more economically disadvantaged and Black students than the comparison group but lower rates of outpatient school-based health claims and psychotherapy visits and higher rates of telehealth mental health visits at baseline. In the empirical models described below, we accordingly adjusted for student demographic composition and economic disadvantage.

Quantitative methods

We estimated AWARE grant effects using two difference-in-differences (DID) methods that leveraged the panel structure of our data and are useful for staggered treatment adoption: the staggered DID approach developed by Callaway and Sant'Anna [19] and a stacked DID method [20]. Each method includes marginally different assumptions, allowing us to assess the sensitivity of results to those assumptions. The staggered DID method assumes parallel trends after conditioning on observed covariates, i.e., in the absence of an AWARE grant, the average outcomes for treated school districts would have evolved similarly to those of the comparison group. It also allows causal effects to be identified even if differences in observed characteristics create nonparallel outcome dynamics between those treated in different time periods. Using this approach, we estimated overall treatment effects compared to districts never treated over the study period, group (cohort) average effects, and partial aggregations to assess how average treatment effects

[4] We do not present analyses including 2021 AWARE grantees, as we only observe them 2 years after grant receipt.

Table 2
Findings on AWARE grant effects from differences-in-differences (DID) estimation

Health and education outcomes	Stacked DID - school level: 2014 and 2019 grantees two pre/four postperiods	Stacked DID- school level: 2014 grantees two pre/ four postperiods	Stacked DID - school level: 2019 grantees two pre/ four postperiods	Staggered DID - district-level: avg. effects across cohorts	Staggered DID - district-level: AWARE cohort-specific ATTs
	AY2012-2023	AY2012-2018	AY2017-2023	AY2012-2023	AY2012-2023
N	15,879	8,073	7,806	1,346	1,346
Mental health diagnoses	**0.02322**	**0.01586**	**0.03052**	**0.00930**	2014: 0.0042 2019: **0.0120**
s.e.	0.00746	0.00715	0.01432	0.00424	
Outpatient mental health visit	0.00029	−0.00402	0.00436	−0.00664	
s.e.	0.00679	0.00655	0.01159	0.00632	
Outpatient school-based claim[a]	**−0.03782**	**−0.03388**	**−0.03916**	**−0.02943**	2014: 0.0111 2019: **-0.0922**
s.e.	0.01103	0.00668	0.01298	0.00505	
Evaluation/ management of mental health diagnosis	0.00159	0.00283	0.00078	−0.00660	2014: −0.0127 2019: **-0.0042**
s.e.	0.00288	0.00401	0.00155	0.00571	
Telehealth - mental health	−0.00424	−0.00172	**−0.01221**	0.00543	
s.e.	0.00269	0.00142	0.00510	0.00323	
Psychotherapy - mental health	−0.00097	−0.00228	0.00128	−0.00229	
s.e.	0.00222	0.00316	0.00162	0.00312	
Mental health medication use	0.00657	0.00303	0.01055	0.00027	
s.e.	0.00453	0.00564	0.00707	0.00497	
Any ED Visit	−0.00768	−0.00226	−0.00980	−0.01870	2014: −0.0115 2019: **-0.0409**
s.e.	0.00755	0.00884	0.01025	0.01376	
Expelled from school	−0.00412	0.00309	−0.01035	**−0.03538**	2014: **-0.0337** 2019: **-0.0439**
s.e.	0.00641	0.00488	0.01195	0.00802	
In-school suspension	−0.00043	0.00000	−0.00148	−0.00451	
s.e.	0.00206	0.00179	0.00387	0.00252	
Out-of-school suspension[a]	**0.01430**	0.00961	**0.02230**	**−0.01470**	2014: −0.0148 2019: **-0.0170**
s.e.	0.00386	0.00566	0.00464	0.00443	

Notes: Standard errors (s.e.) are reported below the coefficient estimates. Coefficients in boldface are statistically significant at $\alpha < 0.05$.
[a] Indicates the parallel trends assumption was not satisfied.

differed by AWARE cohorts. We employed the doubly robust approach with stabilized inverse probability weighting and wild bootstrap standard errors that combines the outcome regression and inverse probability weighting approaches but only requires a correct specification for one of these two, minimizing the consequences of potential model misspecifications.[5] Corrections for multiple hypothesis testing (given the numerous outcomes examined) were generated by the statistical package used in estimation via a multiplier bootstrap procedure to compute critical values of the supremum t-statistic [19,21]. In the stacked DID approach, we constructed separate data sets for each sub-experiment (grant award), vertically combined them to form a stacked analytic file, trimmed to attain balance in the number of preperiods and postperiods for each sub-experiment, and estimated DID regressions to identify average causal effects [20]. A key advantage of this method is the ability to select the number of preintervention and postintervention periods in the analysis, which determines what portion of the weighted effect comes from a given AWARE grant cohort. With data available through academic year 2023, we specified two preintervention and four postperiods, which accords similar weight to the 2014 and 2019 grantees over the study period. The estimates are interpreted as variance-weighted averages of school-year average treatments on the treated (ATTs) for AWARE grantees. Both the staggered DID and stacked DID methods allow for exploration of whether the effects of AWARE grants are heterogeneous across school districts and time periods.

Qualitative analysis

The qualitative data were collected through in-depth interviews with each school district that received an AWARE grant to illuminate grant implementation and effectiveness from the perspective of school district staff in decision-making roles, with most districts interviewed twice. In the qualitative analyses, transcripts and informal reflexive memos were developed for each interview, followed by multiple rounds of coding and sub-coding using NVivo programming software to identify key

[5] More specifically, the estimator is consistent if either the outcome regression model is correctly specified (even if the propensity score model is wrong) or if the propensity score (IPW) model is correctly specified (even if the outcome model is wrong). This works because the augmentation terms cancel the asymptotic bias from the misspecified component. As long as one nuisance estimate is consistent, the key group-time average treatment effect estimate is still consistent, ensuring resilience to misspecification and providing stable inferences [19].

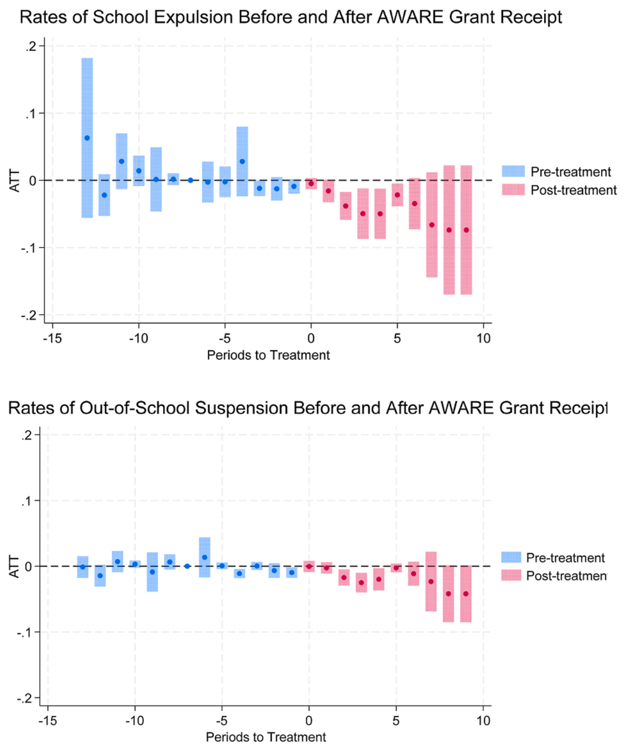

Figure 1. Staggered difference-in-differences estimation of relationship of AWARE grant receipt to rates of exclusionary discipline outcomes.

themes and inform and contextualize the quantitative results on AWARE grant effects. Online Appendix B describes in greater detail the interview procedures, content, transcription, and qualitative coding and analysis.

Results

Quantitative analysis findings

Table 2 presents findings on AWARE grant effects on diagnoses of low-income student mental health conditions, mental health-care utilization, and disciplinary outcomes. The first three columns present results from the school-level stacked DID estimation specifying two pretreatment periods and four post-treatment periods, with the first column including results for the 2014 and 2019 AWARE grantee cohorts combined and the second and third columns including treatment periods specific to the 2014 and 2019 AWARE grantees, respectively. The group (cohort) average treatment effects from the staggered DID district-level estimation are included in the fourth column, and for statistically significant effects (at $\alpha < 0.05$), we also report the separate 2014 and 2019 AWARE grantee effects in the fifth column.

The results show that diagnosed mental health conditions increased after AWARE grants were awarded (compared to non-grantees)—about 1–2 percentage points (ppts), over a baseline mean of about 16%—when examined for both cohorts (from the baseline year to 2023). The results in columns two, three, and five, showing the results separately for the 2014 and 2019 cohorts, indicate that increases in diagnosed mental health conditions were about 1 ppt higher for the 2019 AWARE grant cohort than the combined average effects. The interview data revealed a strong consensus among grantees that these resources were filling critical funding gaps and enabling identification of students with mental health needs, sometimes for the first time.

The DID estimations also consistently associated AWARE grant receipt with 3–4 ppt statistically significant reductions in outpatient school-based claims. In relation to the outpatient school-based claims baseline rate of about 2.0% (already comparatively lower relative to schools without AWARE grants or SBHCs), these associated reductions in claims are sizable. The staggered DID model suggests a substantially larger (9 ppt reduction) decline for the 2019 cohort; however, as indicated in Table 2, the parallel pretrends assumption was not satisfied for all staggered DID estimates for this outcome. In an alternative specification, we restricted the comparison group to rural districts (given that all AWARE grantees were rural), but this did not improve the pretrends comparability in the staggered DID estimation and reduced statistical power; we therefore characterize this specific estimated relationship as an association (vs. effect). Online Appendix C shows that the stacked DID results are not sensitive to the specification of the rural-only comparison group or the district-level unit of analysis.

Other statistically significant findings in Table 2 differ between the stacked and staggered DID estimations and/or appear primarily for the 2019 AWARE grantee cohort. For example, the staggered DID estimates indicate a small (0.4 ppt) reduction in evaluations and management of mental health conditions for the 2019 AWARE cohort, which includes the COVID-19 pandemic period and may be related to the approximate 1 ppt decline observed in tele-mental health visits for 2019 grantees in the stacked DID estimation. In addition, although we observe a negative relationship between AWARE grant receipt and emergency department (ED) visits across the estimations, only the staggered DID estimate is statistically significant (4 ppt reduction) for the 2019 grantees, relative to a baseline ED visit rate over 30%.

The staggered DID analyses also show statistically significant estimated declines in school expulsion rates—a disciplinary action that can last as long as 1 year—of about 3.5 ppts across the 2014 and 2019 AWARE cohorts (with a slightly larger estimated effect for the 2019 grantees), over a baseline rate of approximately 8%. Similarly, the staggered DID results indicate a negative association (about −1.5 ppt) between out-of-school suspension (OSS) rates—an exclusionary disciplinary action that lasts less than 10 consecutive days—and AWARE grant receipt, compared to a baseline rate of about 5%. Figure 1 presents the staggered DID results graphically, showing similar patterns for both of these disciplinary outcomes, which are more precise for the earlier post-treatment periods that include data from both AWARE grant cohorts. However, because the pretrends assumption was not satisfied for the staggered DID out-of-school suspension estimation, and the stacked DID estimates indicate the opposite association, we suggest further exploration of this outcome with additional years of postpandemic data.

Qualitative analysis findings

In interviews conducted with AWARE district staff, we consistently heard that the grants make “a world of difference” in their ability to provide school-based mental health services through direct staff hires and contracting with area mental health organizations. For instance, one grantee described the direct hire of three social workers with the grant funds, as well as a contract with a mental health agency that sends mental health professionals into the schools and then bills insurance. A coordinated school health director in another grantee district noted that they had "[not had] mental health services for a long time," and when three new employees were hired through the AWARE grant, they were immediately overwhelmed with the extent of unmet student mental needs. Similarly, a coordinated school health director in a district with a newly received AWARE grant described discovering very deep level of mental health needs among their students: "we are just scraping the surface" with the grant resources.

A key goal of AWARE grantees was the development of referral pathways for students to access services offered within schools or through community providers. One grantee explained that referrals go first to the school counselor, who then determines whether to refer the student to school-based staff for assistance or out to a community health provider. Because community providers sometimes experience difficulties reaching families after receiving a referral, and families sometimes decline services because of stigma associated with an outside provider, there was a general preference conveyed in our interviews to hire in-school mental health staff. In addition, because of the complexities of billing Medicaid for services to low-income students, district staff appreciated that the grants afforded them the opportunity to offer mental health services without needing to inquire about students' health insurance status or families' ability to pay. These stated preferences for hiring new district mental health employees (vs. contracting for staff whose services would need to be billed) may help to explain the reductions in outpatient school-based claims (on an already low baseline rate) that we observed empirically.

Some AWARE grantees also described how grant funds were frequently used to initially support and "stabilize" the role of community providers in delivering mental health services until they could shift to billing for services provided to students. The grants typically enabled districts to contract with a mental health agency that provided therapists who would come into the schools to serve students, with the grants covering the cost of services for uninsured students and families without the ability to pay.

Reflecting that Tennessee prioritized distressed, rural school districts and each AWARE grantee was located in a rural area, a common theme among the interviewees was how rurality constrained their access to providers. For example, one interviewee lamented: "There's just not enough mental health providers for children or adults in this community… very, very, very limited." Another AWARE grantee commented: "The doctors aren't nearby and don't see the kids in person; they have to get an appointment via telehealth just to get seen." Consequently, developing telehealth capacities in partnership with health-care organizations was often a goal of grantees but also one that they were challenged to sustain. Interviewees described difficulties with adequate technology, training, and having dedicated staff on the school side and providers on the other end. As one interviewee explained: "While we have the telehealth, it's extremely difficult to manage that without extra nurses… It's just hard for them to pause for 30 minutes to support just one kid. So really, you almost would need a full nurse to go around the schools, as they make a [telehealth] appointment."

Lastly, an explicit goal of the AWARE grant program in Tennessee is to reduce the number of youth removed from school for disciplinary issues. AWARE grantees described numerous strategies deployed with grant resources toward this goal. As one interviewee illuminated, "We're building that Tier II service [strategies to support students at risk of or with mild behavioral health challenges]… so that they're not just getting a punishment, but they're actually getting a behavioral intervention, a corrective behavior support." Another interviewee similarly described efforts to shift school culture from punitive to supportive: "So you may come to understand that this kid standing on the desk screaming and not letting you conduct your lesson may be doing that for a reason—they're in pain or something they've experienced; there's strategies to talk a kid down and refer them somewhere… so that kid can come back and stay."

Discussion

Although limited to 5 years, AWARE grants were not intended to be a one-time intervention but rather to facilitate the development of sustainable, holistic, school-based systems of mental health supports. In Tennessee, the grants were allocated to school districts that lacked existing mental health service capacities. Our qualitative findings suggest that newly hired mental health staff subsequently uncovered deep levels of unmet student mental health needs. These insights aligned with our quantitative findings that diagnoses of low-income student mental health conditions increased after AWARE grants were introduced.

As the State of Washington and California evaluations likewise found, AWARE grantees in Tennessee prioritized hiring permanent school-based mental health staff (instead of relying on contracted providers) to avoid a range of challenges experienced when students are referred out for services, including follow-through, stigma, insurance issues, and provider instability (exacerbated by rurality) [9,10]. We see this preferred approach to services delivery as consistent with the decline in outpatient school-based claims observed empirically. AWARE grantees reported that building internal capacities to provide more services "in-house" has "taken off a lot of pressure" from educational staff and allowed them to "look deeper at what we can do" and sustain with school district support.

Lastly, the magnitude of the estimated decline in the rate of exclusionary disciplinary actions associated with AWARE grants (compared to non-AWARE districts) was sizable, following concerted actions by grantee school districts to change school culture and practices on school discipline. The alignment of the quantitative results with these reported efforts to decrease expulsions and out-of-school suspensions may reflect that these grantee efforts were having the intended effect on reducing youth removals from school, similar to what was found in the State of California and West Virginia County Project AWARE evaluations [10,12].

Conclusion

This study adds rigorous empirical evidence to our understanding of effects of federal AWARE grants on district and school-level utilization of mental health services, rates of diagnosed mental health conditions among low-income students, and student disciplinary outcomes. The empirical and qualitative findings suggest that the grants are likely achieving their core objectives of increasing school capacities to identify and better respond to students' mental health needs.

In terms of policy implications, our results suggest that continuing AWARE grant funding and expanding it to more rural, economically distressed school districts with high mental health needs could be beneficial for schools and student well-being. With additional years of data, some promising trends in other outcomes, such as a decline in ED visits, may emerge as grant impacts as well. Yet, no new competition for AWARE grants has been announced in 2025, and other sources of school mental health funding (including from the Bipartisan Safer Communities Act) have recently been cut.

We also reiterate several limitations of this study. We study one of the 49 states that have received AWARE grants (Tennessee) and only the first two rounds of grants. This limits the generalizability of our study findings. In addition, our linked health and education data are for a low-income sample; thus, we would miss any effects for students who never had a Medicaid record (less than a third of Tennessee school-aged students in our study period). Lastly, although we employ two rigorous DID approaches to estimate AWARE grant effects, we recognize that some unobserved factors might have influenced which school districts received the grants, potentially biasing our effect estimates.

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Supplementary Data

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